path—Indira Gandhi International Airport and Safdarjung airport—and the urban sprawl in between. The flight, of course on auto-pilot, will be a closely coordinated sortie—once the drone travels beyond the pilot's visual line of sight, it will need active coordination with the airport's air traffic control (various approvals needed are in place). The trials, involving drone logistics start-up TechEagle, will start next month. The primary aim is to lay down a blueprint for flying drones in such complex scenarios, says Shah. The eventual use-case: ferrying rare blood units and diagnostics samples from the regional hospital to AIIMS New Delhi which has a high-capacity robotic testing laboratory.

LIBERAL POLICY

"Before September 2021, drone delivery was merely a concept, though R&D was active around it," says Ankit Kumar, CEO, Skye Air Mobility. In just under a year, however, there's a rush of activity to explore commercial operations. As to how that happened, here's the long story short: unmanned aerial vehicles for military uses and aeromodelling have been around for years but consumer drones began picking up around 2010 with mostly Chinese-made drones appearing in the market. Following a pizza delivery episode back in 2014 in Mumbai, India's aviation regulator responded with a ban on drones until the government laid down a policy. This eventually came in 2018 but while it allowed manufacture and sale of drones, flying operations remained a grey area.

This chicken-and-egg situation carried on till March 2021 when a fresh set of rules were notified, but the list of mandatory permissions and licences only grew. Finally, in August 2021, this time with the prime minister's office taking keen interest, a new liberalised policy which relaxed most of the restrictions came out. Since then, India's drone ecosystem has taken off.

In quick succession, a slew of policies followed—these included opening up 90 per cent of the Indian airspace as a green zone for flying operations up to a height of 400 ft, online applications via a drone management platform called DigitalSky, guidelines on Unmanned Aircraft System Traffic Management (UTM), a ban on imported drones, a production-linked incentive (PLI) scheme for manufacturers and subsidies for agriculture drones. The government's stated goal is to make India a global drone hub by 2030.

"India was behind the rest of the world in terms of how to operationalise these things," says Ankit Mehta, co-founder of IdeaForge which was an early entrant into the UAV space. "But now we have leapfrogged all of them with a far more implementable regime." By that, he's referring to the calibrated policy steps that many countries had taken.

Still, experiments and specific operations have been going on for a longer period globally than they have in India. For instance, American drone manufacturer Zipline says it has completed over 300,000 commercial deliveries, many of them serving medical facilities in African countries such as Rwanda and Ghana. Besides, when it comes to key components for drones, Indian manufacturers currently rely on imports, a scenario that the PLI scheme hopes to reverse.

EMERGING SECTORS

Mehta reckons the disruption from the pandemic played a key role in pushing the case for using drones—be it monitoring containment zones or delivering medicines to remote places or even tackling a locust swarm, it suddenly became a useful proposition. "It (the pandemic) completely changed the game," he says, "drones started becoming a technology with a million uses." Around the same time, the Survey of Villages Abadi and Mapping with Improvised Technology in Village Areas (Swamitva) scheme was announced. So far, drone surveys of 160,000 villages have been completed.

THE FLYPAST

Drones come in various shapes and sizes now, marked for a myriad uses

CLASSIFICATION OF DRONES

(by maximum all-up weight including payload)



NANO

250 gms or less in weight



MICRO

250 gms to 2 kg



SMALL

2-25 kg



MEDIUM

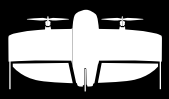
25-150 kg



LARGE

More than 150 kg

DRONE TYPES AND USES



FIXED WING

Land survey and mapping



SINGLE/MULTI ROTOR

Surveillance, photography, inspection, crop spraying, survey and mapping

HYBRID

Combines long range and heavy payload capability for most drone applications